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Introduction

This model analyzes student performance using data from the UC Irvine Machine Learning Repository. The data looks at the demographics of students, their grades, and social and school related features of their lives.The model aims to predict a student's final grade based on the amount of time they spend studying, the amount of absences they have, the number of classes they have failed in the past, and their first and second period grades.

Data Description

The data contains information about secondary school students from two secondary schools in Portugal. In this model we focus on the numerical school related features to see how they impact students' performance. The model looks at the variables study time, failures (number of past class failures), absences (number of times a student has been absent), G1(first period/ first semester grades), and G2(second period/ second semester grades). We look at these to try to predict G3 (Final Grades).

Models and Methods

To predict students' performance I used an OLS(Ordinary Least Squares) regression using stats model to create a linear regression. The data set was divided into training and test sets. The performance of the model was evaluated using MSE(Mean Squared Error), R-squared, and the interpretation of the coefficients. A correlation matrix was also used to see how

Results and Interpretation

The regression shows that G1 and G2 were very strong predictors of a student's final grades. The results of the regression and the correlation matrix, and the coefficients of the variables show that there is a strong positive relationship between the first and second period grades and the final grade. The other variables of the regression have a weak negative correlation with final grades, so while they are significant they don't tell us much about what the final grade of a student would be. The model has a good fit overall with the r-squared being about .85 meaning about 85% of the dependent variable can be explained by the predictor.

Conclusion

The model shows that students' final grades can be predicted based on their earlier grades and other academic variables. The next steps could be expanding the model by adding in categorical variables, along with social variables to improve the models predictions.